

Practice Question

SUBMISSION INSTRUCTIONS (LMS)

1. Complete the assessment within the strictly enforced 30-minute time limit.
2. Create and run your shell script (sys_snapshot.sh) on your assigned Linux terminal environment.
3. **Once completed, upload the following TWO items to the Course LMS submission portal:**
 - Item A: Your completed script file (sys_snapshot.sh).
 - Item B: A single text/log output file or screenshot showing the successful execution and verified results in the staging directory.

Task Scenario: System Snapshot & Log Management

You are working as a junior Linux system administrator. You are required to write a Bash script named **sys_snapshot.sh** in your home directory (~). The script must automate directory management, process logging, file moving/copying, and system permission configuration.

1: Directory Setup & Inspection

- Create a directory named ~/snapshot_workspace if it does not already exist.
- Create a sub-directory named ~/snapshot_workspace/staging.
- List all contents of your home directory using detailed/long-format flags and save the output to ~/snapshot_workspace/workspace_contents.txt.

2: System Process Monitoring

- Capture a single-batch snapshot of active system processes
- Filter/redirect the output to save only the top 10 lines into ~/snapshot_workspace/process_log.txt.

3: File Management & Operations

- Copy process_log.txt from ~/snapshot_workspace/ to ~/snapshot_workspace/staging/ using the cp command.
- Move workspace_contents.txt from ~/snapshot_workspace/ into ~/snapshot_workspace/staging/ using the mv command, renaming it to archived_contents.txt in the process.

4: Permissions Setup (10 Marks)

- Make your script file executable using the chmod command.
- Set permissions on ~/snapshot_workspace/staging/archived_contents.txt so that it is strictly Read-Only for all users using chmod.

Step 5: Execution & Verification (10 Marks)

- Execute your script sys_snapshot.sh.
- Verify that the staging directory contains both archived_contents.txt and process_log.txt with correct permissions.